Utilizing Copper Compounds to Develop an Anti-Cancer Agent

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Abstract

Lung cancer is a leading cause of death throughout the world. According to the American Cancer Society (ACS), an estimated 124,990 deaths from lung cancer are expected in the U.S. in 2026. Most current treatments work by triggering apoptosis, which is a built-in process that causes cells to die. Over time, cancer cells can become resistant to this, making them harder to kill. Ferroptosis is a different way that cells can die — it happens when iron causes a buildup of toxic lipid molecules inside the cell. Recent research has shown that copper may also play a role in this process, which could make treatments more effective. In the present study, we tested the efficacy of copper compounds towards cancer cell properties including cell viability, cell migration, and clonogenic ability. Furthermore, we investigated the effects of combining known ferroptosis inducers (Erastin and RSL3) with copper compounds on the expression of different proteins involved in stress-associated signaling. Our results showed that combining Erastin with a copper compound greatly reduced cancer cell growth and cell viability, while having little effect on normal cells. Interestingly, the combined treatment of ferroptosis inducers and copper compounds led to greater activation of cell stress and death pathways compared to either compound alone. These findings could help develop new treatments for cancers that have become resistant to standard therapies.

Keywords: ferroptosis, cuproptosis, copper compounds, lung cancer, cell viability, Western blot, apoptosis resistance

Introduction

Cancer is a major threat to human health. In 2025, over 2 million new cancer cases were predicted in the United States, which works out to about 4–5 new cases every minute (6). Most cancer treatments, like chemotherapy, work by triggering apoptosis — a process where the cell essentially destroys itself. The problem is that cancer cells can develop resistance to apoptosis over time, making the treatment less effective (9). Because of this, researchers have been looking for other ways to kill cancer cells. One promising method is ferroptosis, which is a type of cell death caused by a buildup of iron-dependent lipid peroxides inside the cell (9).

Iron levels in the body are mostly controlled by how much is absorbed (8). Copper is a similar metal and has been found to affect the ferroptosis process. A study by Wang and colleagues (5) found that combining a ferroptosis inducer with a copper compound increased copper-related cell death (called cuproptosis) in liver cancer cells. They used qRT-PCR and Western blotting to show that the drugs Erastin and sorafenib could trigger both ferroptosis and cuproptosis at the same time (5). Another study found that Erastin combined with a compound called Celastrol could kill cancer cells without being toxic at normal doses (4). Liu and colleagues (4) also confirmed that RSL3 and Erastin are both strong inducers of ferroptosis, and that other chemicals can make this process happen faster.

A copper compound called CuATSM has been shown to increase ferroptosis in brain cells, but not in cancer cells (2). A similar compound, CC101, had never been tested in cancer cells before this study (12; 11). Research by Iakovidis and colleagues (10) showed that copper and related compounds can affect many different pathways inside cells that are linked to cancer. Wang and colleagues (7) also found that ferroptosis and cuproptosis may work together and could be combined to kill cancer cells more effectively.

Research Question

Does combining copper compounds (CC101 and CuATSM) with ferroptosis inducers (Erastin and RSL3) reduce lung cancer cell growth more than using either compound by itself, and does this combination leave normal lung cells unaffected?

Hypothesis

It is hypothesized that the combination treatment of ferroptosis inducers (Erastin or RSL3) with copper compounds (CC101 or CuATSM) will cause a greater decrease in lung cancer cell viability, colony formation, and migration compared to using either compound alone. The combination treatment is also expected to cause a greater increase in proteins related to cell stress and cell death pathways.

Methodology

For all assays, 250,000 cells were plated for each cell line: H2030, HPL1D, A549, and H358. After plating, cells were counted and split into treatment groups.

Cell Viability Assay

For the cell viability assay, 1,000 cells per well were plated in six replicates in 96-well plates and allowed to sit overnight so they could attach. The next day, cells were treated with either vehicle alone (control), a copper compound alone (CC101 or CuATSM), a ferroptosis inducer alone (Erastin or RSL3), or one of the following combination treatments: CC101+Erastin, CuATSM+Erastin, CC101+RSL3, or CuATSM+RSL3. Cells were then incubated for 48 hours at 37°C with 5% CO_2. After 48 hours, cell viability was measured using Alamar Blue following the manufacturer's instructions (3). Absorbance values were recorded and used to calculate the percent of cells still alive compared to the control group.

Colony Formation Assay

For the colony formation assay, 1,000 cells per well were plated in triplicate in 6-well plates and allowed to sit overnight to attach. Cells were then treated with either vehicle alone (control), a copper compound alone (CC101 or CuATSM), Erastin alone, or a combination treatment (CC101+Erastin or CuATSM+Erastin). Cells were allowed to grow for 10 days, with media replaced as needed. After 10 days, the media was removed, colonies were fixed using 70% methanol, and stained with Coomassie Blue. The number of colonies in each well was then counted and recorded.

Cell Migration Assay (Scratch/Wound Healing)

For the cell migration assay, 200,000 cells per well were plated in triplicate in 6-well plates for each cell line and allowed to grow until the well was fully covered (confluent), which took about 24 hours. A pipette tip was then used to scrape a straight line through the middle of each well,

creating a gap to simulate a wound. Each well was photographed right after the scratch was made (0 hr). Cells were then treated with either vehicle alone (control), a copper compound alone (CC101 or CuATSM), Erastin alone, or a combination treatment (CC101+Erastin or CuATSM+Erastin). Photos were taken again at 24 hr and 48 hr after the scratch. The amount of gap closure in each photo was measured and used to calculate the percent of wound closure compared to the 0 hr image.

Western Blot Analysis

For the Western blot, H2030 and H358 lung cancer cells were plated on 60 mm dishes and allowed to sit overnight to attach. Cells were then treated with either vehicle alone (control), Erastin alone, a copper compound alone (CC101 or CuATSM), or a combination treatment (CC101+Erastin or CuATSM+Erastin). After 24 hours, cells were collected and broken open using lysis buffer to extract the proteins. Protein concentration was measured using the BCA Protein Assay Kit. From each sample, 40 µg of protein was mixed with SDS loading buffer and boiled, then run through a 4–12% SDS-PAGE gel to separate the proteins by size. The separated proteins were transferred onto a membrane and probed with primary antibodies targeting ER stress proteins and cell death proteins. The bands were made visible using SuperSignal™ West Femto reagent. The membrane was then stripped and reprobed with a GAPDH antibody as a loading control. Band brightness was measured using densitometry software, and each band's intensity was divided by the GAPDH band from the same lane to normalize the results.

Data Analysis

Results were compared between groups using an unpaired Student's t-test. A difference was considered statistically significant when $P < 0.05$ ($^{*}P < 0.05$; $^{**}P < 0.005$). All data were recorded and graphed in spreadsheet software. Each experiment was done in triplicate or more unless stated otherwise.

Results

The figures below show the data collected from each experiment. Analysis and explanation of these results are in the Discussion section.

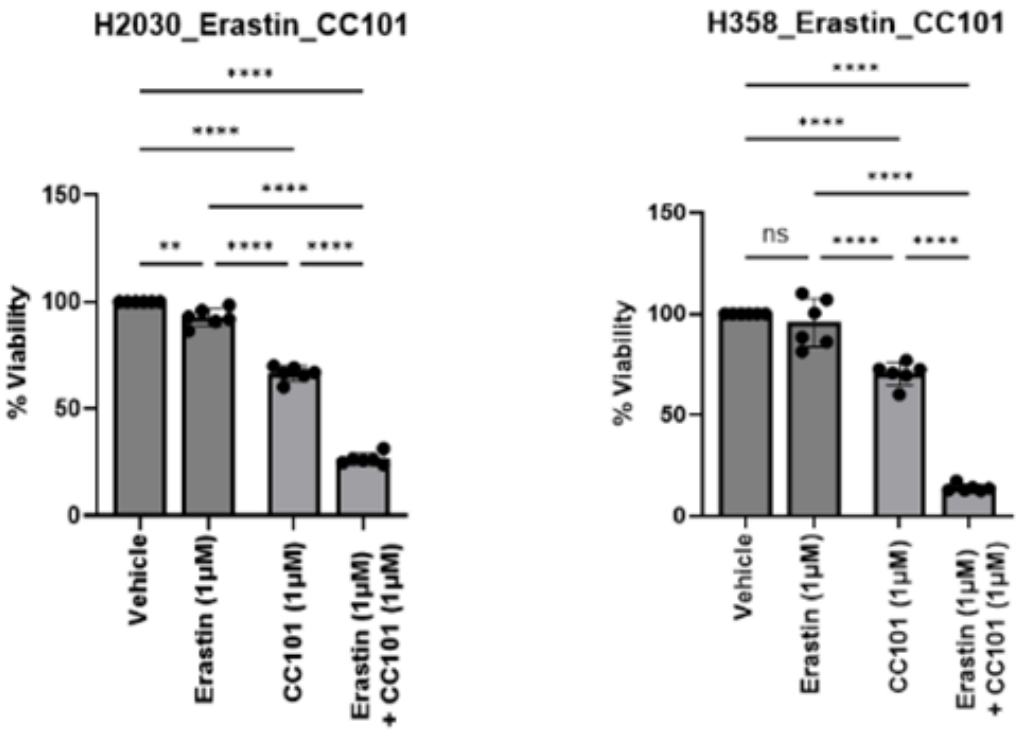


Figure 1: **Figure 1A. Cell Viability Assay — CC101 and CC101+Erastin.** Cells were treated with CC101 alone or with CC101 and Erastin together. Viability is shown as a percent of the vehicle control. $^{*}P < 0.05$; $^{**}P < 0.005$.

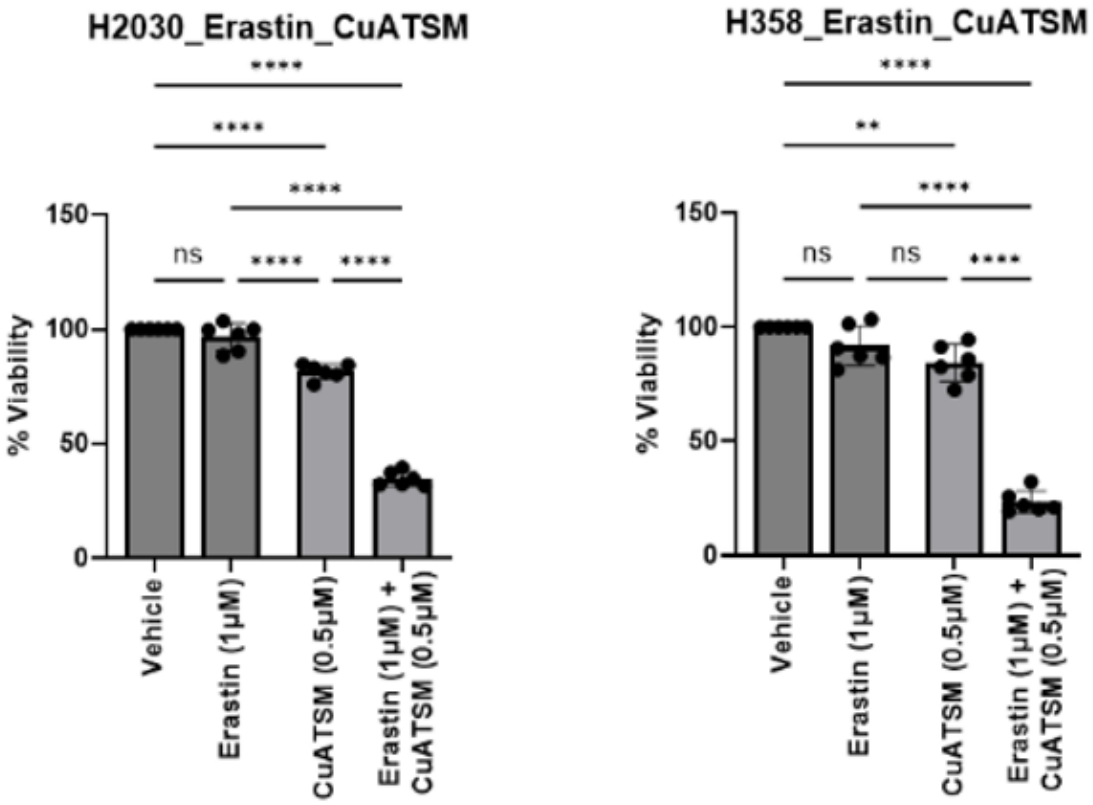


Figure 2: **Figure 1B. Cell Viability Assay — CuATSM and CuATSM+Erastin.** Cells were treated with CuATSM alone or with CuATSM and Erastin together.

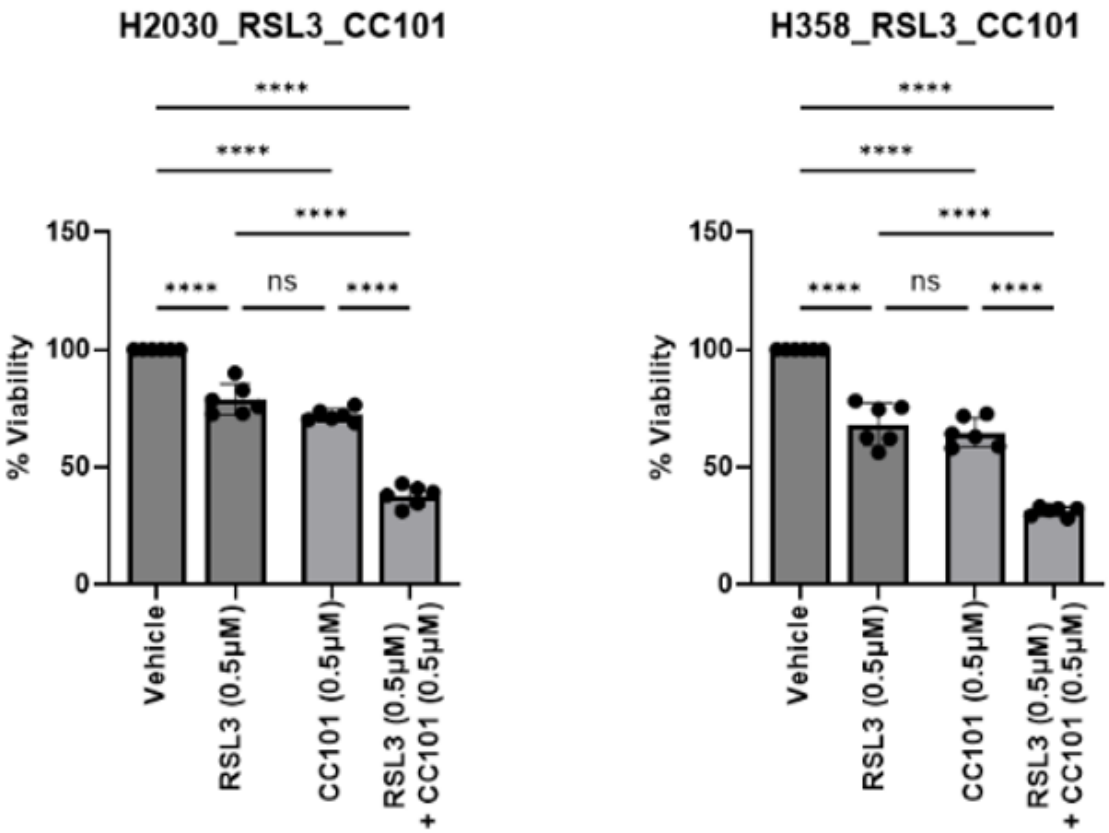


Figure 3: **Figure 1C. Cell Viability Assay — CC101 and CC101+RSL3.** Cells were treated with vehicle alone, CC101 alone, RSL3 alone, or CC101 and RSL3 together.

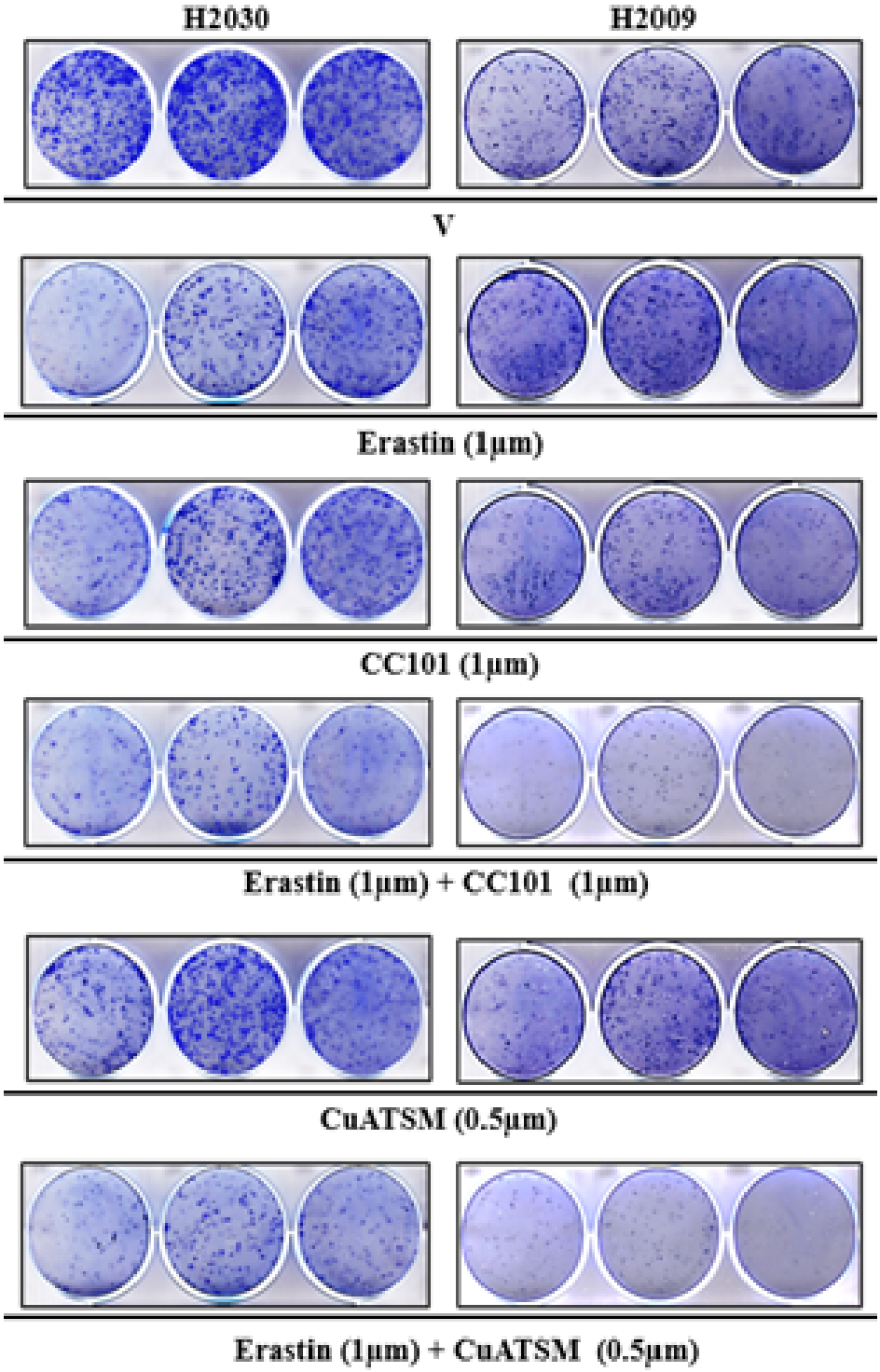


Figure 4: **Figure 2. Colony Formation Assay.** Colonies were stained with Coomassie Blue after 10 days of growth. Cells were treated with CC101, CuATSM, Erastin, or a combination. Colony counts were compared to the vehicle control.

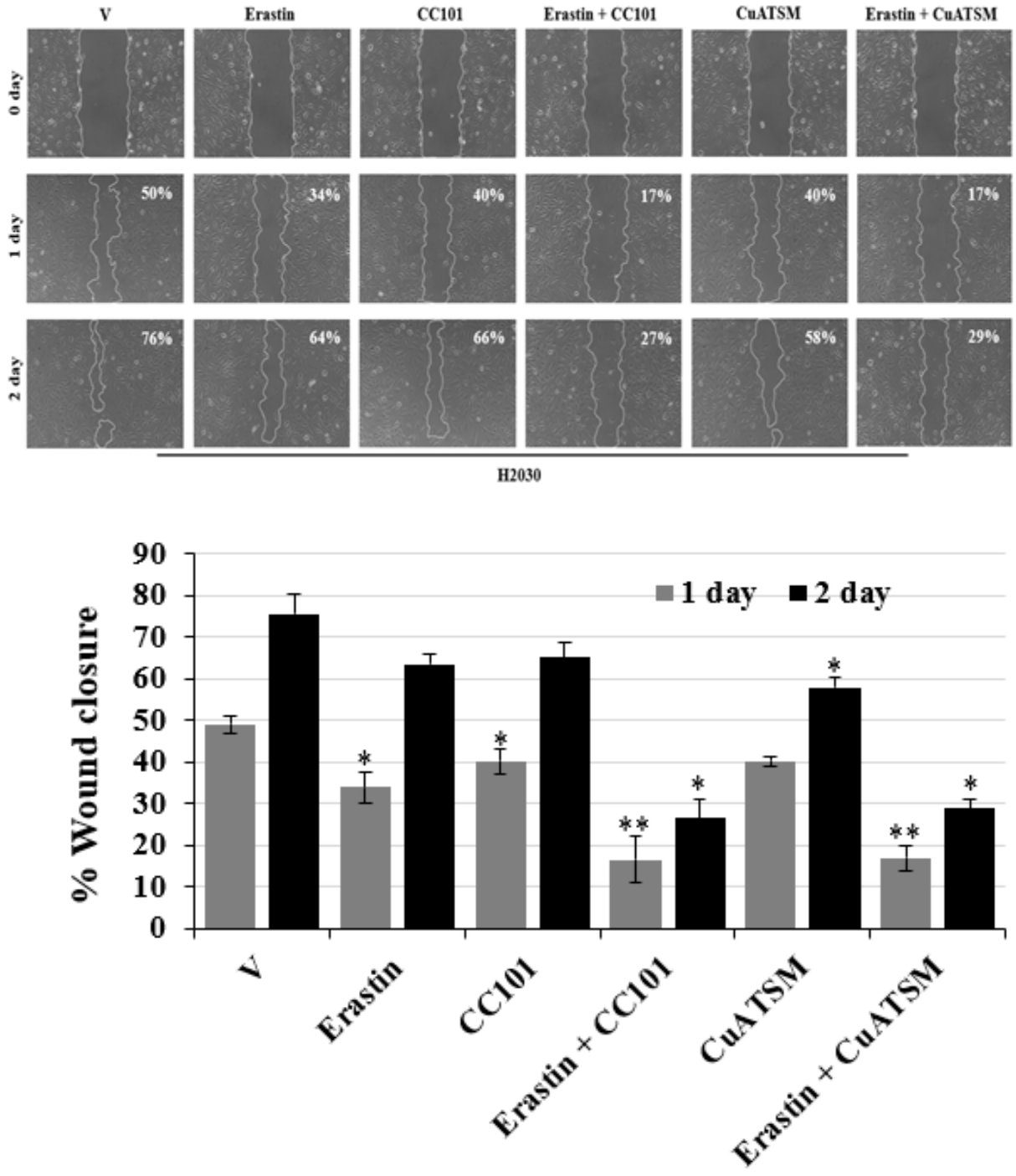


Figure 5: **Figure 3. Cell Migration Assay — Scratch Wound Photos.** H2030 cells were treated with CC101, CuATSM, Erastin, or a combination. Photos were taken at 0 hr, 24 hr, and 48 hr after the scratch.

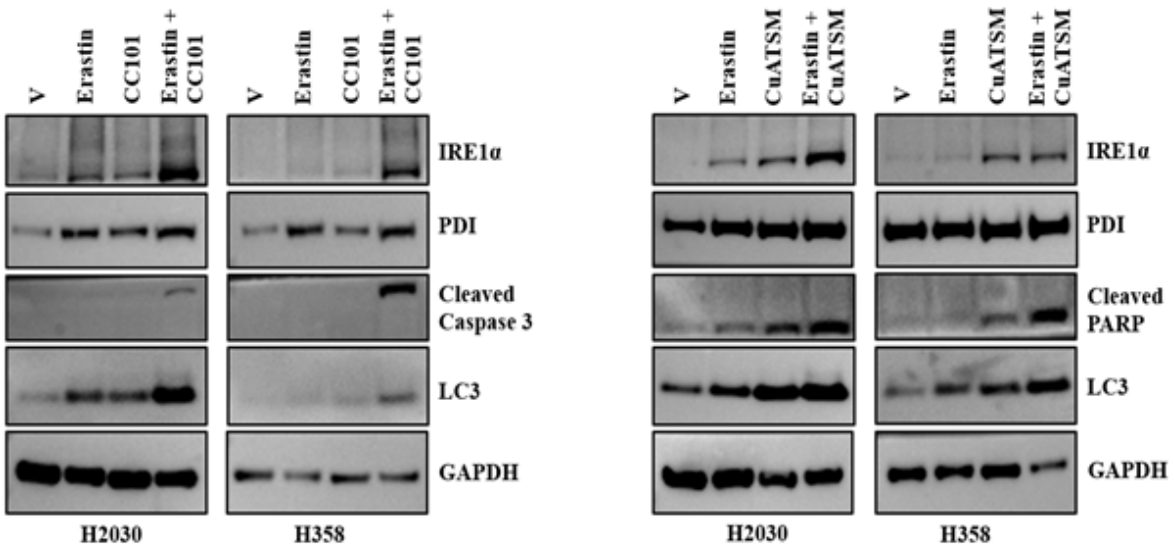


Figure 6: **Figure 4. Western Blot — ER Stress and Cell Death Proteins.** H2030 and H358 cells were treated with vehicle, Erastin, copper compounds, or a combination for 24 hours. Protein was then extracted and analyzed using Western blot. GAPDH was used as a loading control.

Discussion

Cell Viability

The cell viability results showed that CC101 and CuATSM both reduced how many cancer cells survived when used on their own. When either copper compound was combined with Erastin, the drop in cell survival was even greater (Figures 1A, 1B, 1C). This suggests that the two types of compounds are working through related or overlapping pathways to kill cancer cells. An important finding was that the combination treatment had little to no effect on the normal lung cells (HPL1D), which shows that it may be targeting cancer cells more specifically.

Colony Formation

The colony formation results matched what was seen in the viability assay. Each individual treatment (CC101, CuATSM, or Erastin alone) caused fewer colonies to form compared to the control. But when the compounds were combined, the number of colonies dropped even more (Figure 2). This means the cells are less able to survive and keep dividing over a longer period when both compounds are used together. This matches what Wang and colleagues (5) found when combining ferroptosis inducers with copper compounds in liver cancer.

Cell Migration

In the scratch assay, all treatments slowed down how fast cells moved to close the wound. The combination treatments caused the biggest slowdown at both the 24 hr and 48 hr time points (Figure 3). This is important because cancer cells that move less are less likely to spread to other parts of the body.

Western Blot Analysis

The Western blot results showed that single treatments caused only small changes in protein levels. The combination treatments caused much bigger changes, especially in proteins related to ER

stress and cell death, in both H2030 and H358 cells (Figure 4). This matches what Iakovidis and colleagues (10) found — that copper compounds can affect multiple cell death pathways at once. Having both ER stress and cell death pathways activated at the same time could explain why the combination treatment works so much better than either compound alone. Wang and colleagues (7) described how ferroptosis and cuproptosis can work together, which likely explains the stronger response seen here.

Conclusion

Ferroptosis is a type of cell death that depends on iron and involves a buildup of toxic lipid molecules (9). Copper has been found to play a role in this process (1). This study tested two copper compounds — CC101 and CuATSM — to see if they could make the ferroptosis inducer Erastin more effective against lung cancer cells.

The results supported the hypothesis. Using Erastin together with either CC101 or CuATSM caused a greater reduction in cancer cell survival, colony formation, and migration compared to using any one compound alone. This was true across multiple lung cancer cell lines (A549, H2030, H358). Normal lung cells (HPL1D) were not significantly affected by the combination treatment, which is an important finding for safety.

The Western blot results showed that the combination treatment activated both ER stress and cell death pathways more than any single treatment. This shows that the two compounds work better together, likely because they hit different but connected pathways (7). This is especially useful for cancers that have stopped responding to standard treatments, since ferroptosis and cuproptosis offer a different way to kill those resistant cells (11; 12).

Future Research

Based on the results of this study, there are several directions that future research could take:

- **Test more drugs:** Try other copper compounds and ferroptosis inducers, including RSL3 combined with copper compounds, to find the most effective combinations.
- **Look at the mechanism more closely:** Measure things like ROS levels, lipid peroxide buildup, and more ferroptosis/cuproptosis proteins to better understand why the combination works so well.
- **Run dose-response experiments:** Test different concentrations of each drug to find the best dose and calculate IC_{50} values.
- **Test in animals:** Move from cell culture to mouse models to see if the treatment works in a living body and to check for any side effects.
- **Test other cancer types:** See if the combination treatment works on other cancers besides lung cancer, especially ones that are resistant to standard treatments.
- **Study why normal cells are unaffected:** Find out what makes normal HPL1D cells less affected by the treatment, which could help explain how to make the therapy even more targeted.

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